

EXPERIMENT 16

Aim

To perform edge detection on an image using the Canny Edge Detection method in Python with OpenCV.

Procedure

1. Import the required libraries, cv2 and numpy.
2. Read the input image from the specified path.
3. Convert the input image from RGB/BGR to grayscale.
4. Apply Gaussian blur to reduce noise.
5. Apply the Canny edge detection algorithm using suitable threshold values.
6. Display the original image and detected edges.
7. Save the edge-detected image.

Code

```
import cv2

# Read the input image
img = cv2.imread("image.jpg")

# Check whether image is loaded
if img is None:
    print("Error: Image not found")
    exit()

# Convert image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```
# Reduce noise using Gaussian Blur
blur = cv2.GaussianBlur(gray, (5, 5), 0)

# Apply Canny Edge Detection
edges = cv2.Canny(blur, 50, 150)

# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Canny Edge Image", edges)

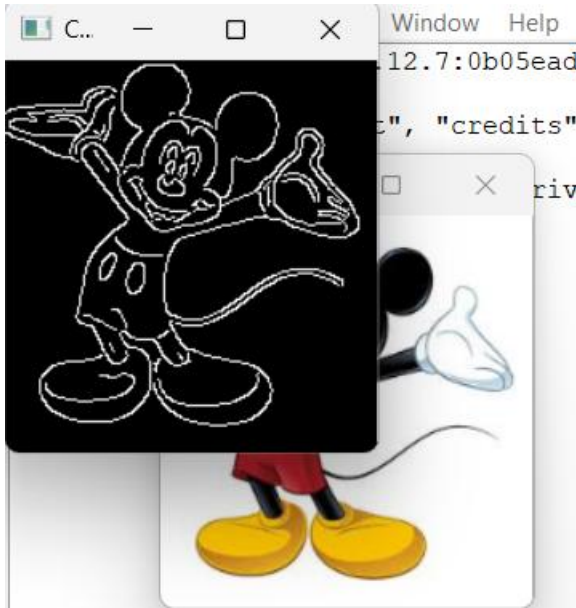
# Save the output
cv2.imwrite("canny_output.jpg", edges)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, Canny Edge Detection was successfully performed on the input image, and the edges of objects were detected and displayed.